Heat is not temperature

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish temperature from energy transferred because of a temperature difference.

SUCCESS CRITERIA - I can...

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “heat” and “temperature” to compare your model or result. Sentence starters are optional.

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

thermal energy: internal energy associated with microscopic motion/interactions

01 Bring back an earlier idea

A toy model requires 15 J to escape, but an object has only 12 J and no later energy input. Does the model predict escape?
Explain.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $Q = mc\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

Worked reasoning: A cup and a bathtub can both be at 40°C. Equal temperature does not imply equal internal energy: amount and type of material also matter.

Your turn: Draw equal temperature indicators but different amounts of water.
Predict heat-flow direction if the water samples are brought into thermal contact at the same temperature.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

03 Look and explain

A cup and a bathtub contain the same substance at 40°C. Are their temperatures equal? Must their total internal energies be equal?

Sentence starter: I notice ... . This supports ... because ... .



Cup: 40°C



Tub: 40°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Build the solution

A warm metal block at 60°C touches water at 20°C.
Draw the initial heat-transfer direction.
Predict whether each temperature rises or falls.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Choose a method

A wheel rolls down a ramp, losing 24 J gravitational energy. Its center-of-mass kinetic energy gain is 20 J with negligible losses. What energy term is missing?

Word help: negligible = small enough to leave out of this model

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

State one way temperature and heat are different, including their usual units.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A small metal spoon is hotter than a large bath. Can temperature alone tell which contains more internal energy?
Explain what else you would need.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3